Volumes and octahedral chains of ideal right-angled hyperbolic four-polytopes

Coxeter Polytopes Research Project

18 October 2026

Abstract

The regular ideal right-angled 24-cell has volume $4\pi^2/3$ and is the unique minimum-volume entirely ideal right-angled hyperbolic four-polytope, by Kolpakov. We prove that the next distinct volume is $8\pi^2/3$ and that its unique equality case is the reflection double of the 24-cell across one octahedral facet. We also prove a structure theorem without a volume assumption: if every facet is a regular ideal octahedron or its triangular-face double, the polytope splits along actual octahedral sections into a chain of regular 24-cells, with opposite cuts in every intermediate cell. The equality theorem uses a separate defect argument to establish this facet hypothesis. The proof combines cusp geometry, a finite small-facet classification, an elementary closed-surface argument, and Lorentz normal identities. The restricted structure theorem does not assert a decomposition or volume divisibility for arbitrary ideal right-angled four-polytopes.

1 Statements and proof dependencies

A polytope in this paper is convex, finite-sided, four-dimensional, and of finite hyperbolic volume. It is *entirely ideal* if every vertex lies at infinity, and *right-angled* if every dihedral angle is $\pi/2$. Write $\mathcal{C}$ for the regular ideal right-angled 24-cell, $\mathcal{O}$ for its regular ideal octahedral facet, and $\mathcal{D}$ for the union of $\mathcal{O}$ with its reflection across a triangular face. The polyhedron $\mathcal{D}$ has nine vertices, eight triangular faces and three quadrilateral faces.

Theorem 1.1 (The published minimum)**.** *For every entirely ideal right-angled hyperbolic four-polytope P,*

$$\mathrm{Vol}(P) \geq \frac{4\pi^2}{3},$$

with equality exactly when P is isometric to $\mathcal{C}$.

This is Kolpakov's theorem [1, Theorem 3]. We recall the volume and incidence formulas underlying it below. Our later all-octahedral recognition argument also recovers its equality statement once those formulas are given.

Theorem 1.2 (The second volume and its equality case)**.** *If P is entirely ideal and right-angled and $P \ncong \mathcal{C}$, then*

$$\mathrm{Vol}(P) \geq \frac{8\pi^2}{3}.$$

Equality holds exactly for the union of $\mathcal{C}$ and its reflection across an octahedral facet. Its f-vector is $(42, 168, 164, 38)$.

The next theorem concerns a restricted class, with no assumed volume. Its additional hypothesis is

$$\text{every facet of } P \text{ is isometric to } \mathcal{O} \text{ or } \mathcal{D}. \tag{1}$$

The word "isometric" fixes the intrinsic geometry, not just the numbers of faces. Equivalently one may assume the corresponding combinatorial types and use three-dimensional right-angled rigidity.

Theorem 1.3 (Octahedral cuts and chains)**.** *Suppose that P satisfies (1). Finitely many pairwise disjoint regular ideal octahedral sections cut P into regular ideal right-angled 24-cells. If P has N facets of type $\mathcal{D}$, then $N = 8k$ for an integer $k \geq 0$, and there are $k+1$ cells. Their adjacency graph is a path. Two cuts incident to one cell use opposite facets. Within this class the isometry type of an existing polytope is determined by its number of cells, and*

$$\mathrm{Vol}(P) = (k+1)\frac{4\pi^2}{3}.$$

We classify existing polytopes under (1); the theorem does not assert existence for every prescribed path length.

The logical order matters. First we prove the gap by facet defects and an injection of opposite cusp facets. Next, at volume $8\pi^2/3$, we prove independently that exactly eight facets are $\mathcal{D}$ and all others are $\mathcal{O}$. Only then do we apply the general chain theorem: $N = 8$ gives two cells and identifies the double. In particular, the facet hypothesis is not inferred from the desired decomposition.

1.1 Published inputs

Kolpakov's Lemmas 1–2 give the incidence and volume formulas used below. Proposition 5 states that an ideal right-angled three-polyhedron with a k-gonal face has at least $2k$ vertices, with equality exactly for the k-antiprism. Section 4.3, Case D and Figure 6 give the unique admissible nine-vertex case. The equality reduction also uses the ten-vertex octahedrite in Figure 7: among the ten-vertex types with only triangles and quadrilaterals it is the unique admissible one, and the edge-adjacency graph of its four quadrilaterals consists of two disjoint edges. The pentagonal antiprism is treated separately. We use the standard right-angled rigidity in dimension three to identify these combinatorial models geometrically, as in Kolpakov's use of Andreev's theorem.

These are three-dimensional small-facet inputs. No classification of arbitrary four-polytopes is used. Opposite-cusp injection, equality-spectrum reduction, closed surfaces, actual cutting sections and the chain theorem are proved here.

2 Discrete volume and facet defect

Write v, e, r, f for the numbers of vertices, edges, ridges and facets of P. An ideal vertex link is a Euclidean rectangular three-box. Counting its vertices and edges and using Euler's identity gives

$$e = 4v, \qquad r = 3v + f, \qquad \sum_{R \in \Omega_2(P)} k(R) = 12v,$$

where $k(R)$ is the number of vertices of a ridge R. Define

$$W(P) = \sum_R (k(R) - 3).$$

It follows that

$$W(P) = 3(v - f) \in 3\mathbb{Z}_{\geq 0}, \qquad \mathrm{Vol}(P) = \frac{\pi^2}{3}(v - f + 4) = \frac{4\pi^2}{3} + \frac{\pi^2}{9}W(P). \tag{2}$$

The latter identity is Kolpakov's volume formula; equivalently, Gauss–Bonnet applied to the reflection orbifold uses $\chi = 1 - f/2 + r/4 - e/8 = (v - f + 4)/4$.

Every facet F is an ideal right-angled three-polyhedron. Its ideal vertices have valency four, so, writing v_F for their number, it has $2v_F$ edges and $v_F + 2$ polygonal faces. Its defect is

$$d(F) := v_F - 6 = \sum_{R \subset F} (k(R) - 3), \qquad \sum_F d(F) = 2W(P). \tag{3}$$

Defect zero characterizes the regular ideal octahedron. A positive defect facet has a nontriangular face of size $k \geq 4$; the antiprism bound $v_F \geq 2k$ implies $d(F) \geq 2$. In particular defect one does not occur.

3 Opposite cusp facets

For a facet F, let $q(F)$ be its number of nontriangular faces and $s(F)$ the number of its cusps whose intrinsic horospherical rectangle is nonsquare. The latter condition does not depend on horosphere scale.

Lemma 3.1 (Opposite-cusp injection)**.** *Suppose the smallest positive facet defect of P is m, and F has positive defect. There are $q(F) + s(F)$ pairwise distinct positive-defect facets other than F: its neighbors across nontriangular ridges, and the facets opposite F at its nonsquare cusps. Therefore*

$$2W(P) \geq d(F) + m\big(q(F) + s(F)\big). \tag{4}$$

In particular, when $d(F) = m$, $2W(P) \geq m(1 + q(F) + s(F))$.

Proof. A neighbor across a nontriangle contains that same nontriangle and is not octahedral. Distinct ridges of F have distinct neighboring facets, since the intersection of two facets is a face.

At a cusp x of F, the face of the rectangular three-box opposite the F-face belongs to a facet F_x. The two rectangles are congruent. If the rectangle of F is nonsquare, so is that of F_x, which cannot be the regular ideal octahedron.

The supporting hyperplanes of F and F_x are asymptotic and meet at exactly the ideal point x. Hence the same F_x cannot occur at a second cusp of F, nor can it meet F across any ridge. Thus all facets in the two lists are distinct and differ from F. Each has defect at least m. Summing their defects in (3) proves the assertion. □

If $P \ncong \mathcal{C}$, a positive-defect facet exists: otherwise $W = 0$ in (3), and the volume formula and Kolpakov's unique minimum identify P with $\mathcal{C}$.

4 The two cusp-shape calculations

Lemma 4.1. *The cusp rectangles of the ideal right-angled k-antiprism have aspect ratio $2\cos(\pi/k)$. Thus for $k \geq 4$ all $2k$ cusps are nonsquare.*

Proof. Set $\theta = \pi/k$, $c = \cos\theta$, and take positive square roots in

$$h^2 = \frac{1-c}{1+c}, \quad r^2 = \frac{2c}{1+c}, \quad b^2 = \frac{2}{c(1+c)}, \quad \mu^2 = \frac{2(1-c)}{c}.$$

Use Lorentz form $(-1,1,1,1)$. The two vertex rings have heights $\pm h$, radius r, and angular offset θ. At $u = (1, r\cos\theta, r\sin\theta, h)$, the four incident normals are

$$N = (h,0,0,1), \quad M_0 = (1,b,0,h), \quad M_2 = (1,b\cos 2\theta, b\sin 2\theta, h), \quad L = (1,b\cos\theta, b\sin\theta, -h).$$

Their squares are r^2, μ^2, μ^2, μ^2. Adjacent products vanish; the opposite pairs are parallel. The identities $bc = r$ and $\mu = 2h/r$ give

$$\frac{M_0 + M_2}{\mu} = \frac{2}{\mu}u, \qquad \frac{N}{r} + \frac{L}{\mu} = \left(\frac{h}{r} + \frac{1}{\mu}\right)u.$$

For two opposite unit normals with sum κu, their separation on a fixed horosphere is proportional to κ. To see this, send u to infinity in the upper-half-space model; the mirrors become parallel vertical planes with unit horizontal normals. The two separations therefore have ratio

$$\frac{2/\mu}{h/r + 1/\mu} = \frac{2r^2}{1+h^2} = 2c.$$

The antiprism symmetries are vertex-transitive. Finally $2c > 1$ for $k \geq 4$ (and equals one for $k = 3$). □

Lemma 4.2. *An ideal right-angled three-polyhedron with nine vertices has $d = 3$, $q = 3$ and $s = 3$.*

Proof. The antiprism bound excludes pentagons and larger faces. The complete published nine-vertex octahedrite list consists of one admissible type: two ideal regular octahedra glued across a triangular face [1, Section 4.3, Case D2 and Figure 6]. Rigidity makes this identification geometric. The three gluing cusps have rectangles formed by joining two squares, hence aspect two. The other six cusps remain square. Its three nontriangular faces are quadrilaterals. □

5 Universal exclusion of intermediate volumes

Lemma 5.1. *A positive-defect facet has at least two nontriangular faces. If it has exactly two, it is an antiprism.*

Proof. If its only nontriangle has size k, then $d = k - 3$, whereas the antiprism bound gives $d \geq 2(k-3)$, a contradiction. If the two sizes are $k \geq l \geq 4$, then

$$d = (k-3) + (l-3) \geq 2(k-3).$$

Thus $k = l$ and equality holds in the antiprism bound; its equality statement identifies the facet with A_k. □

Proof of the lower bound in Theorem 1.2. Choose a facet F of smallest positive defect $m \geq 2$. Lemma 3.1 gives $2W \geq m(1+q+s)$. The following cases are exhaustive.

1. If $F = A_k$, then $m = 2k-6$, $q = 2$, and $s = 2k = m+6$ by Lemma 4.1. Hence $2W \geq m(m+9) \geq 22$.

2. A non-antiprism with $m = 2$ is impossible: $v_F = 8$ gives $k \leq 4$ for every face, so its defect is two quadrilaterals and Lemma 5.1 applies.

3. If $m = 3$, Lemma 4.2 gives $2W \geq 3(1 + 3 + 3) = 21$.

4. If $m = 4$, every face has size at most five. A pentagon gives equality in the antiprism bound and has already been treated. Otherwise there are four quadrilaterals, so $2W \geq 4(1+4) = 20$.

5. If $m \geq 5$ and F is not an antiprism, Lemma 5.1 gives $q \geq 3$, whence $2W \geq 4m \geq 20$.

Thus $W \geq 10$. Since W is divisible by three, $W \geq 12$. Equation (2) proves the lower bound. In particular the three prospective intermediate defects $3, 6, 9$, with respective volumes $5\pi^2/3$, $2\pi^2$ and $7\pi^2/3$, cannot occur. □

6 The intrinsic doubled octahedron

The three vertices of the gluing face of $\mathcal{D}$ have a simple intrinsic description: their horospherical rectangles are nonsquare. At each of these vertices the rectangle is obtained by adjoining two unit squares along one side, and therefore has aspect ratio two. The remaining six rectangles are square. The three quadrilateral faces of $\mathcal{D}$ meet pairwise at its three nonsquare cusps and have no common edges. At a nonsquare cusp, the two quadrilateral sides are the two long sides of its 2 by 1 rectangle.

The convex hull of the three nonsquare cusps is an ideal triangle T_F whose relative interior lies in the relative interior of a doubled facet F. This is its original gluing triangle. Each side of T_F is the interior diagonal of a quadrilateral face of F. At a nonsquare cusp, T_F meets the horospherical rectangle along the transverse segment halfway along its long direction. Figure 1 records these facts in the Euclidean link, where their metric meaning is literal.

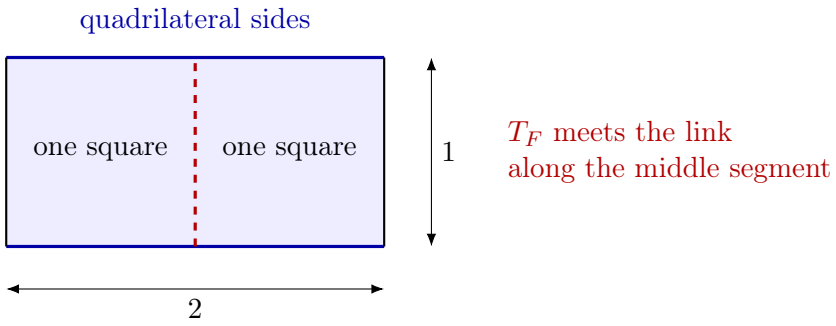


Figure 1: The cusp rectangle of a doubled octahedron at a gluing vertex. The blue sides belong to its quadrilateral faces. The red segment is the link of the internal triangle. This is an exact Euclidean link diagram, not a perspective drawing of $\mathbb{H}^4$.

At a square cusp of $\mathcal{D}$, exactly one incident polygonal face is a quadrilateral. At a cusp of $\mathcal{O}$, none is. These facts will be used both in the equality reduction and in the general octahedral cutting argument.

7 The facet spectrum at the second volume

In this section alone assume $\mathrm{Vol}(P) = 8\pi^2/3$. Then $W = 12$ and $\sum_F d(F) = 24$. A facet is *positive* when its defect is positive; all others are regular ideal octahedra. Let N be the number of

positive facets and m their minimum defect. For any positive facet, opposite-cusp injection gives the nonminimal form

$$24 \geq d(F) + m\big(q(F) + s(F)\big). \tag{5}$$

We shall prove the following statement before making any use of Theorem 1.3.

Proposition 7.1. *There are exactly eight positive facets, each isometric to $\mathcal{D}$. All other facets are isometric to $\mathcal{O}$.*

For continuity with the small-facet list, we also write D9 for $\mathcal{D}$. Its three quadrilateral faces have no common edges and meet pairwise at its three nonsquare cusps. Every triangle contains at most one of those cusps; the quadrilaterals contain two each, as opposite vertices. At a square cusp exactly one incident face is quadrilateral. These are the intrinsic facts of Section 6.

7.1 Acute intersections

We use the following projection to pass from pairwise facet adjacency to actual common faces. Give every supporting wall a unit outward spacelike normal. Distinct wall products are nonpositive for an acute Coxeter chamber. If $n_1, \ldots, n_k$ are mutually orthogonal and p is a strict future timelike interior point, put

$$u = p - \sum_{i=1}^{k} \langle p, n_i \rangle n_i.$$

Then $u^2 = p^2 - \sum_i \langle p, n_i \rangle^2 < 0$, $\langle u, n_i \rangle = 0$, and every other wall product is strictly negative: each subtracted product $\langle p, n_i \rangle \langle n_i, n_j \rangle$ is nonnegative. The projected timelike vector has the future sign, for example because $\langle p, u \rangle = u^2 < 0$. Thus the k walls meet in an actual codimension-k face. Three mutually adjacent right-angled facets meet along an actual edge; four would give a finite vertex, forbidden here.

Consequently, if F has nontriangular neighbors G and H and G is adjacent to H anywhere in P, the corresponding two faces of F share an edge. This excludes an adjacency elsewhere that is invisible in the intrinsic facet picture.

7.2 The minimum positive defect is three

No antiprism A_k with $k \geq 5$. Its defect is $d = 2k - 6 \geq 4$, $q = 2$ and $s = 2k = d + 6$. Even the universal $m \geq 2$ in (5) gives

$$24 \geq d + 2(d + 8) = 3d + 16 \geq 28,$$

a contradiction.

No defect two facet. Such a facet is the square antiprism A4. It has eight nonsquare cusps and two quadrilateral faces; each cusp belongs to exactly one of these quadrilaterals. If one face of a rectangular three-box is nonsquare, at least one of the other two opposite-face pairs is nonsquare. Otherwise both ratios to the third side would be one and all sides would be equal. Hence at each cusp of A4 at least two of its four ridge-neighbor facets are nonoctahedral. At most one is the unique quadrilateral neighbor through that cusp. At least one triangular neighbor is therefore positive. Covering the eight cusps by triangular faces requires at least three distinct such triangular neighbors. These are distinct from F, its two quadrilateral neighbors, and its eight nonsquare-cusp opposites, by the opposite-cusp injection. Thus at least 14 positive facets exist, giving total defect at least 28. This is impossible. We have $m \geq 3$.

The branch $m \geq 5$ is impossible. There are at most four positive facets. Every positive facet is a non-antiprism, so has at least three nontriangular faces. Their neighbors must be distinct positive facets. Thus there are exactly four, all pairwise adjacent. The acute-intersection fact would give a finite vertex.

The branch $m = 4$ is impossible. There are at most six positive facets. Choose a defect four facet F. A pentagon would force A5, already excluded. Thus F is the Figure 7 ten-vertex octahedrite, with four quadrilaterals paired by edge adjacency. Let $G_1, \ldots, G_4$ be their four positive neighbors. The induced adjacency graph on these four neighbors is contained in that matching, by the acute-intersection fact. Each G_i has at most F, its one matching partner and one further positive facet as possible nontriangular neighbors. Every G_i needs at least three such neighbors, so a further facet must exist and $q(G_i) = 3$. None of the G_i can have defect four, since that case has four nontriangular faces; hence each has defect at least 5. Including F and the necessary further facet already gives $4 + 4 \cdot 5 + 4 = 28$, a contradiction.

It follows that $m = 3$, so some facet is D9.

7.3 Exactly eight positive facets

A D9 facet and its three quadrilateral neighbors and three nonsquare-cusp opposites give seven positive facets. Since $m = 3$ and the total defect is 24,

$$7 \leq N \leq 8.$$

If $N = 8$ all defects are three, as desired. Suppose $N = 7$. For every D9 facet, the seven just counted exhaust all positive facets. In particular:

> Under $N = 7$, no triangular ridge of a D9 facet has a positive neighbor.

The seven defects are at least three and sum to 24. The possible excess partitions above seven copies of three are: one defect six; a defect five and a defect four; or three defects four. There are at most three facets whose defects exceed three.

There can be no defect four facet F. Its four quadrilaterals occur in edge-adjacent pairs. Suppose one quadrilateral neighbor G were D9, and let H be the neighbor across its paired quadrilateral. The edge shared by the two quadrilaterals of F is an actual edge shared by F,G,H. In G, the faces cut out by F and H share this edge. The former is a quadrilateral. The latter cannot be a quadrilateral because D9's quadrilaterals share no edges, and cannot be a triangle because H is positive and $N = 7$ forbids a positive triangular neighbor. Thus every one of F's four quadrilateral neighbors would have defect greater than three. There are at most two such facets other than F, a contradiction.

We are left with one defect six facet F and six D9 facets. Every nontriangle of F is shared with a D9, and is therefore a quadrilateral. There are exactly six quadrilaterals. No two share an edge, by the same argument with their common D9 neighbor: two adjacent quadrilaterals of F would force either adjacent quadrilaterals or a positive triangular neighbor in D9.

The facet F has twelve vertices. Its six quadrilaterals give24 quadrilateral-vertex incidences. At a four-valent vertex no two consecutive faces can both be quadrilaterals; therefore there are at most two quadrilaterals there. Equality in the incidence count forces exactly two, alternating with two triangles, at every vertex. Form a graph J whose vertices are the six quadrilateral faces and whose edges are the twelve vertices of F, joining the two incident quadrilaterals. It is a simple 4-regular graph on six vertices: two distinct quadrilateral faces cannot share two vertices without sharing an edge, since their intersection is a face of the convex three-polyhedron. Its complement is a perfect matching, so it is the octahedral graph.

The embedding and medial recovery are explicit here. Draw each new edge from a center of one quadrilateral to their shared original vertex and then to the center of the other quadrilateral. Alternation at every original vertex makes this a spherical embedding, with the eight original triangular regions as its eight triangular faces. Those face cycles are distinct: a connected embedded graph with more than three vertices cannot have both sides of one 3-cycle as faces, since then no other vertex or edge could lie on either side. The octahedral graph has exactly eight 3-cycles, so all its 3-cycles are precisely these faces. This determines the face incidences, not merely the abstract graph. Recover F with one vertex for each edge of J, its six quadrilaterals around the six vertices of J, and its eight triangles around those eight 3-cycles. This is exactly the cuboctahedral face lattice. No general embedding-rigidity theorem or unbounded facet census is being assumed.

The ideal right-angled cuboctahedron has all twelve cusp rectangles nonsquare, of aspect $\sqrt{2}$. An exact model in Lorentz coordinates has vertices

$$(1, x), \quad x \text{ a permutation of } (\pm 1, \pm 1, 0)/\sqrt{2};$$

its six square-wall normals are $(1, \pm\sqrt{2}e_i)$, of norm one, and its eight triangle-wall normals are $(1, (\pm 1, \pm 1, \pm 1)/\sqrt{2})$, of norm $1/2$. At $(1, 1/\sqrt{2}, 1/\sqrt{2}, 0)$, the two opposite square normals sum to $(2, \sqrt{2}, \sqrt{2}, 0)$, and the two opposite *unnormalized* triangle normals have the same sum. Unit normalization multiplies the latter by $\sqrt{2}$. The horospherical width ratio is therefore $\sqrt{2}$, and symmetry gives it at every vertex. The model is the cube $|X_i| \leq 1$ truncated by $\sum |X_i| \leq 2$ in scaled Klein coordinates, so its combinatorics are explicit. Right-angled rigidity identifies F with this model.

The opposite-cusp injection for F would now give $1 + 6 + 12 = 19$ positive facets, contradicting $N = 7$. We conclude:

$$\boxed{N = 8, \quad d(F) = 3 \text{ for every positive facet.}}$$

All other facets are regular ideal octahedra. All nontriangular ridges are quadrilaterals; there are twelve of them.

8 Cusp boxes under the octahedral facet hypothesis

We now drop the volume assumption of Section 7 and assume only (1). In particular W need not be12 and the number of doubled facets need not be eight. We prove Theorem 1.3 directly from this intrinsic facet geometry.

Lemma 8.1. *Every noncubic cusp box of P has side proportions $2 : 1 : 1$. Its four nonsquare faces belong to doubled facets and form a belt; its two square end faces belong to ordinary octahedral facets. Every adjacency within the belt is a quadrilateral ridge of P.*

Proof. Each face rectangle of a cusp box is intrinsic to a facet of P. Its aspect ratio is therefore one or two. If the three side lengths were distinct, the ratio of the largest to the smallest would exceed two, since the two consecutive unequal ratios must both be two. Thus a noncubic box has proportions $2 : 1 : 1$ or $2 : 2 : 1$.

Consider a $2 : 2 : 1$ box. The four nonsquare faces must be doubled facets. In each, the two long sides are quadrilateral sides. Each square end face consequently has four incident quadrilateral sides. An octahedral cusp has none, and a square cusp of a doubled facet has exactly one. Neither permitted facet type can supply an end face. The $2 : 2 : 1$ box is impossible.

In a $2:1:1$ box, the four long rectangles are the belt. Their mutual sides are the long sides and hence are quadrilateral ridges. The square ends meet the belt only along its short, triangular sides. All four incidences at an end are triangular, so the end cannot be a square cusp of a doubled facet. It is octahedral. □

9 The internal triangles form closed surfaces

Join two doubled facets when they share a quadrilateral ridge. The distinguished triangles in these facets have the same internal diagonal in that ridge, so we glue them along this diagonal. The gluing is geometrically defined; there is no choice of an abstract edge identification.

Lemma 9.1. *After adjoining their ideal vertices, the glued triangles form a finite closed two-manifold. Its connected components are exactly the components of the quadrilateral adjacency graph of doubled facets. Every edge lies in two triangles and every vertex lies in four.*

Proof. The interior of a distinguished triangle lies in the relative interior of its facet. Interiors of different facets are disjoint. An edge of the triangle is in the relative interior of an actual quadrilateral ridge, which lies in precisely two facets. Thus each edge has two incident triangles and has a disk neighborhood at each of its finite points.

It remains to check the ideal vertices. By Lemma 8.1, the triangles incident to one such point come from the four doubled facets in a single $2:1:1$ belt. Their four middle segments join end to end around the box, giving the simple square shown in Figure 2. They do not cross and do not leave an open end. This square is the link of the point in the glued surface. Coning the link gives a disk after the ideal point is adjoined. There are no other triangle vertices.

All four doubled facets at such a vertex are already connected through their quadrilateral belt. Therefore identifying ideal vertices does not join two different components of the quadrilateral adjacency graph. The components of the graph and of the surface coincide. □

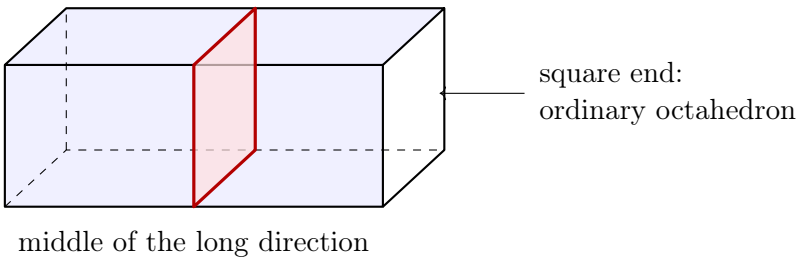


Figure 2: Four internal triangles meet an ambient $2:1:1$ cusp box in its transverse middle square. Its boundary is the four-segment link used to prove that the compactified triangle complex is a two-manifold. The filled square indicates the transverse section; only its boundary lies on the box surface.

Lemma 9.2. *A connected component containing n triangles contains exactly eight triangles, twelve edges and six vertices.*

Proof. Write e and v for its edge and vertex counts. The incidence counts of Lemma 9.1 give

$$e = \frac{3n}{2}, \qquad v = \frac{3n}{4}, \qquad \chi = v - e + n = \frac{n}{4}.$$

We shall use only the elementary inequality $\chi \leq 2$ for a connected closed cellulated surface; we do not assume the surface is a sphere. Here is a direct proof in the present setting. Choose a spanning tree A of its one-skeleton. Delete from the dual graph the edges dual to A. The remaining dual graph is connected. Otherwise, the boundary between one nonempty proper set of triangular faces and its complement would be a nonempty subgraph of A. At each vertex its degree is even: a walk around the circular link enters and leaves the chosen set equally often. This gives a nonempty Eulerian subgraph of a tree, which is impossible. We may therefore choose a dual spanning tree disjoint from A. The two trees use $v-1$ and $n-1$ distinct edges, so $e \geq v+n-2$ and $\chi \leq 2$. It follows that $n \leq 8$, and the integer $\chi = n/4$ also shows that n is divisible by four.

We next obtain a lower bound within the same component. Choose one doubled facet F. Its three quadrilateral faces have three distinct doubled neighbors. Its three nonsquare cusps also have three opposite doubled facets. The latter are distinct from one another and from every ridge neighbor: their walls are asymptotic to the wall of F at exactly one specified ideal point, whereas a ridge neighbor intersects its wall along a plane. These opposite facets belong to the same component, because opposite belt members are connected by the four-cycle of quadrilateral adjacencies. Together with F we have found seven distinct facets in the component. Thus $7 \leq n \leq 8$ and $4 \mid n$, forcing $n = 8$. The values of e and v follow. □

10 From six cusps to eight wall normals

Fix one component. Make a graph J whose six vertices are its cusps and whose edges are the twelve internal diagonals. Every vertex has degree four.

Lemma 10.1. *The graph J is the octahedral graph. Its eight triangles are the eight distinguished facet triangles. The doubled facets can therefore be indexed by $s \in \{\pm 1\}^3$ so that quadrilateral adjacency changes exactly one sign.*

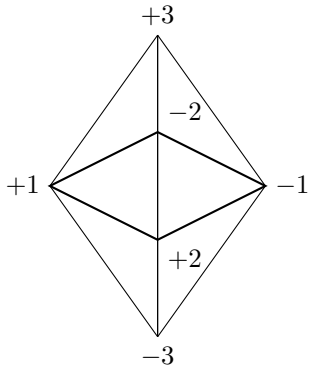
Proof. First, J has no parallel edges. If two different quadrilateral ridges joined the same two cusps, their incident facets would include at least three of the four belt members at either cusp. Among three belt members there are two opposite ones. Their walls are asymptotic at that cusp and cannot contain the second cusp. This is a contradiction. Two distinct ridges cannot have the same pair of incident facets, since the intersection of two convex facets is one face. Thus J is simple.

A simple four-regular graph on six vertices has complement a perfect matching, so it is the octahedral graph. Each distinguished triangle gives a three-cycle in J. Two facets cannot give the same three-cycle: its geometric triangle would then have an interior point in the relative interiors of two different facets. The octahedral graph has exactly eight three-cycles, obtained by choosing one vertex from each of its three opposite pairs. All eight occur. Label these opposite pairs by $1, 2, 3$, with a sign on each member. A triangle, and hence a doubled facet, is indexed by its three choices of sign. Two triangles share an edge exactly when one sign changes. □

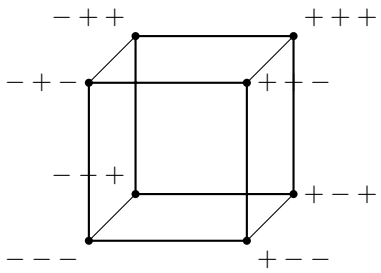
11 Equal widths put the triangles in one hyperplane

We work in $\mathbb{R}^{4,1}$ with form $\langle x, x\rangle = -x_0^2 + x_1^2 + \cdots + x_4^2$. A wall has an outward unit spacelike normal n, and its halfspace is $\langle x, n\rangle \leq 0$. Let n_s be the normals of the eight doubled facets just indexed.

Lemma 11.1. *On each square of the cube in Figure 3, the sums of the two opposite pairs of normals agree.*



cusps and diagonals



doubled facets and quadrilaterals

Figure 3: The two incidence graphs. The six-cusp graph is octahedral; its eight triangles index the doubled facets. The dual cube records quadrilateral adjacency. The signs specify which cusp from each opposite pair belongs to a distinguished triangle. These are incidence drawings, not metric projections of the polytope.

Proof. The square is a cusp belt. Its opposite wall pairs bound the two short directions of the $2:1:1$ box, so their Euclidean separations on a common horosphere agree. The use of unit normals matters. With cusp null vector $\xi = (1,0,0,0,1)$, the vertical wall $x_i = a$ in upper-half-space has unit normal (a, e_i, a). The opposite wall $x_i = b$, with the other outward orientation, has normal $(-b, -e_i, -b)$. Their sum is $(a-b)\xi$. Thus its coefficient is the separation, with the same normalization for both pairs. Equal widths give equal sums. □

Lemma 11.2. *There are mutually orthogonal vectors a, b_1, b_2, b_3 such that*

$$n_s = a + s_1 b_1 + s_2 b_2 + s_3 b_3, \qquad a^2 = -\tfrac{1}{2}, \quad b_i^2 = \tfrac{1}{2}.$$

Proof. The equations of Lemma 11.1 say that every mixed second finite difference on the cube vanishes. Expanding $s \mapsto n_s$ in sign characters leaves only its constant and three linear terms, giving the displayed affine expression. Expanding the eight equations $n_s^2 = 1$ and comparing sign characters gives

$$\langle a, b_i \rangle = 0, \quad \langle b_i, b_j \rangle = 0 \ (i \neq j), \quad a^2 + b_1^2 + b_2^2 + b_3^2 = 1.$$

Adjacent facets meet orthogonally. If s' differs from s only in coordinate i, then $0 = \langle n_s, n_{s'} \rangle = 1 - 2b_i^2$. Thus every $b_i^2 = 1/2$, and the remaining identity gives $a^2 = -1/2$. □

The span U of these four vectors is Lorentzian and has dimension four. Its orthogonal complement is a spacelike line. The intersection of U with $\mathbb{H}^4$ is a genuine hyperbolic hyperplane H, orthogonal to all eight doubled-facet walls. In orthonormal coordinates on U, the wall normals are $(1, s_1, s_2, s_3)/\sqrt{2}$, with the future sign determined by their positive cusp-null sums. Their halfspaces cut out

$$S = \{(t, x) \in \mathbb{H}^3 : |x_1| + |x_2| + |x_3| \leq t\}, \tag{6}$$

the regular ideal octahedron. Its six null rays are $a \pm b_i$. They are exactly the actual cusps of the component, because the opposite belt-normal sums are $2(a \pm b_i)$.

Convexity now supplies the step that an abstract graph cannot: since these six ideal points belong to the closure of P, their hyperbolic convex hull S is contained in P. Conversely the eight actual facet inequalities imply $P \cap H \subset S$. Therefore

$$P \cap H = S.$$

Each face of S is its facet's distinguished internal triangle. Each edge of S is the internal diagonal of an actual quadrilateral ridge, as illustrated in Figure 4.

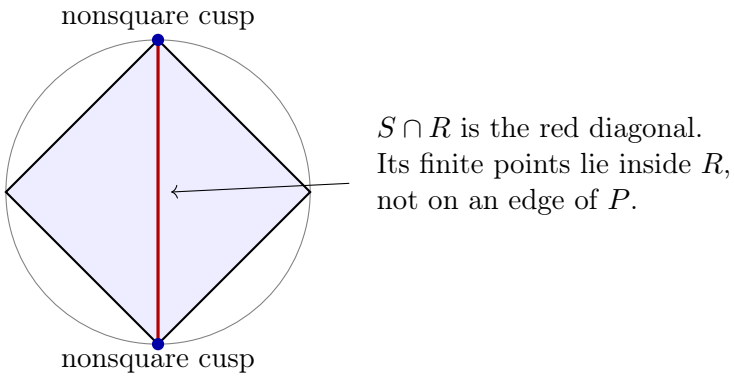


Figure 4: A quadrilateral ridge R in the Klein disk. Its two nonsquare cusps are opposite vertices. Their diagonal is an edge of the cutting octahedron. This distinction is what prevents an unrelated facet from touching the section at a finite edge point.

12 Why the hyperplane is a legitimate cut

Lemma 12.1. *No facet outside the chosen eight meets S at a finite point. Cutting along H gives two finite-volume entirely ideal right-angled four-polytopes. The eight doubled facets become ordinary octahedra; every other facet is uncut.*

Proof. Let F be any outside facet. Its supporting functional is nonpositive on S. If it vanishes at a relative interior point of a face of S, it vanishes on that face. A finite point of S lies in its interior, in a triangular face interior, or in an edge interior. A triangular face interior belongs to the relative interior of one of the chosen doubled facets, so it cannot belong to F. An edge interior lies in the relative interior of a ridge whose only two incident facets are the chosen doubled facets. It cannot belong to F either. A supporting functional vanishing at an interior point would contain all of S, contradicting the face-interior case. Thus F has no finite intersection with S.

If F crossed H, a finite segment in F would meet $P \cap H = S$ at a finite point. Consequently no outside facet crosses. The chosen eight facets are cut along their genuine gluing triangles into octahedra. The center of S is strict on all eight wall inequalities and, by what was just proved, on every other wall inequality. It is an interior point of P; both halves have nonempty interior.

In the Klein model, a vertex of a half lying on H must be a vertex of S. Otherwise it lies in a nontrivial segment of S and is not extreme. Away from H, a vertex is an old vertex of P. The only vertices of S are its six old ideal cusps, so no finite vertex is introduced. The new facet S meets only the eight cut walls, all orthogonal to H. Every new dihedral angle is right; all other angles are inherited. Finite volume follows by containment in P. □

The facet spectrum (1) is preserved by the cut. There are exactly eight fewer doubled facets, counted over both pieces. Repeating the construction therefore terminates.

There is also a useful geometric disjointness statement. Every outside doubled facet lies on one side of the first cut. Two quadrilateral-adjacent outside facets cannot lie on opposite sides, since their shared ridge would give a finite intersection with S. Thus a whole other component lies on one side. Its section, being the convex hull of its six cusps, lies on that side as well. A noncubic

cusp belongs to only one belt component and has octahedral ends; two sections cannot share an ideal vertex. The outside-facet argument excludes finite boundary contact. Hence the sections are pairwise disjoint, including their ideal endpoints.

13 Recognizing a piece with only octahedral facets

We give a direct recognition argument rather than using a four-dimensional classification as a final black box.

Lemma 13.1. *An entirely ideal right-angled four-polytope whose facets are all isometric to $\mathcal{O}$ is isometric to $\mathcal{C}$.*

Proof. Place one facet F on a facet of a model regular 24-cell. Its eight triangular faces agree with those of the model facet. For any one of these triangles, the adjacent supporting wall is uniquely determined by the triangle and orthogonality to F's wall: its outward normal is the intrinsic outward unit normal of that triangle within F's wall.

In this adjacent wall, its octahedral facet is determined by the shared ideal triangle and the side on which it lies. To justify this last assertion, take any isometry between two regular ideal octahedra with that triangular face. Every isometry of the face extends to a symmetry of the octahedron, so the isometry can be adjusted to fix the face vertices pointwise. An isometry of $\mathbb{H}^3$ fixing three distinct ideal points is the identity or reflection in their plane; the specified side selects the identity. The neighboring facet is therefore the whole model octahedron, not only a piece on the same wall.

Repeat across its faces. The facet adjacency graph of the model 24-cell is connected. The propagation matches all 24 model facets, with their full vertex sets, to actual facets of the piece. Their halfspaces put the piece inside the model cell. Conversely the closure of the piece contains every ideal model vertex, and convexity contains their hull, the model cell. The two polytopes agree. □

Each final piece is consequently a 24-cell. If the original polytope has N doubled facets, Lemma 9.2 gives $N = 8k$. Each of the k cuts increases the number of pieces by one, so there are $k+1$ cells. Their adjacency graph is a tree: starting with one piece, each separating cut replaces one vertex of the current adjacency tree by two vertices joined by one edge.

14 Why the tree is a chain

The restriction on facet types has one more consequence. In the Lorentz model a regular 24-cell has outward unit normals

$$(1, r), \qquad r \in \{(\pm 1, \pm 1, 0, 0) \text{ and all coordinate permutations}\}.$$

The spatial vectors are the roots of D_4. Distinct roots have dot products $-2, -1, 0, 1$, with -2 only for opposite roots.

Lemma 14.1. *If a cell in the decomposition meets two cutting sections, these are opposite facets of that cell.*

Proof. The sections have disjoint closures. If their roots are r_1, r_2, then $r_1 \cdot r_2$ is negative: dot product one gives an actual common triangular ridge, and dot product zero gives a common ideal vertex. Thus the dot product is -1 or -2.

Suppose it is -1. The vector $r_1 + r_2$ is another D_4 root, and its cell facet Q is orthogonal to both cut walls, since

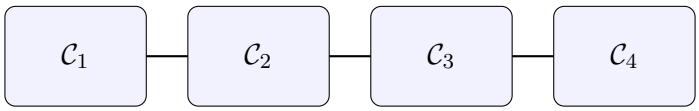
$$-1 + (r_1 + r_2) \cdot r_i = 0 \qquad (i = 1, 2).$$

It cannot itself be a cutting facet, since it meets the two sections along triangles. It is part of an original facet of P. Across each of the two cuts there is a neighboring regular 24-cell. Its corresponding facet is the reflection of Q and lies in the same original supporting wall, because that wall is orthogonal to the cut. This neighboring facet cannot be another internal cut: it meets the first cut along a triangle, whereas all cuts have disjoint closures. Thus the original facet contains three regular ideal octahedra.

Their relative interiors are disjoint. If two had an open overlap, a point of that overlap avoiding every other cell wall would have an inward normal collar contained in both cells. Their four-dimensional interiors would overlap, which is impossible for pieces obtained by cuts. The original facet has intrinsic volume at least three times $\mathrm{Vol}(\mathcal{O})$; it can be neither $\mathcal{O}$ nor $\mathcal{D}$. This contradicts (1). Hence the dot product is -2, and the two facets are opposite. □

No cell can have three pairwise opposite facets. The adjacency tree has maximum degree two and is a path. Once an end cell and its first gluing facet are placed, every next cell is its reflection across the gluing facet, and the next gluing facet in an intermediate cell is uniquely the opposite one.

For completeness, there is no independent gluing twist. The regular 24-cell is flag-transitive. Given an isometry of an octahedral facet, send one facet flag to its image by a cell symmetry. This symmetry stabilizes the facet and has the required restriction, because an octahedral isometry fixing a flag is the identity. Thus every facet symmetry extends to the cell. Two cells on opposite sides of the same facet are therefore uniquely placed up to the ambient reflection. This proves the isometry assertion in Theorem 1.3.



two cuts in an intermediate cell
must use opposite octahedral facets

Figure 5: The resulting adjacency graph is a path. The boxes denote whole 24-cells and are not metric four-dimensional drawings. The opposition condition comes from the common-neighbor argument in Lemma 14.1, rather than from the drawing.

15 Volume, face counts, and the limit of the result

The regular ideal 24-cell has volume $4\pi^2/3$. Volume additivity proves the formula in Theorem 1.3. The common defect identities also give $2W = 3N = 24k$, consistently with (2).

If there are $m = k + 1$ cells, the complete face counts are

$$(v, e, r, f) = (18m + 6,\ 72m + 24,\ 68m + 28,\ 14m + 10).$$

Indeed each gluing identifies six ideal vertices and removes two facets while merging eight neighboring pairs. There are no extra vertex identifications between nonadjacent cells: a common cusp would lie on every separating cut along their path, whereas different sections have no common ideal

vertex. The edge and ridge formulas then follow from the incidence identities above. There are $8(m-1)$ doubled facets, $6m+18$ ordinary facets and $12(m-1)$ quadrilateral ridges. All other ridges are triangles. For $m=2$ this recovers $(42, 168, 164, 38)$.

These formulas do not establish the same quantization for arbitrary ideal right-angled four-polytopes. Without (1), a facet may have more complicated cusp rectangles, the internal triangles used here may not exist, and the local links need not be the four-segment belts of Lemma 9.1. A successful extension must replace those geometric facts, not merely repeat the Euler count.

16 Equality and attainment

Equality in Theorem 1.2. If $\mathrm{Vol}(P) = 8\pi^2/3$, Proposition 7.1 puts P in the Oct/D class and gives $N = 8$. Theorem 1.3 therefore gives $k = 1$: there are exactly two regular 24-cells joined across one octahedral facet. The placement and no-twist argument in the chain proof shows that they are mutual reflections in that facet. Conversely their reflection double is an entirely ideal right-angled polytope of twice the 24-cell volume, as the explicit construction below verifies. This proves the equality assertion. □

16.1 An exact model of the attaining double

In Lorentz form $(-1,1,1,1,1)$, take the 24 unit normals $(1,a)$, where a runs through the D_4 roots having exactly two nonzero entries, each ±1. In the time-one Klein section their ideal vertices are the eight $\pm e_i$ and the sixteen $(\pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2},\pm\frac{1}{2})$. The inequalities are $|x_i|+|x_j|\le 1$; they are bounded and their origin is strict. Here is a finite exact verification of the vertex list. For each of the $\binom{24}{4} = 10,626$ four-element subsets of the walls, form the 4×4 matrix of their spatial coefficients. If its rank is four, solve the four affine equalities over $\mathbb{Q}$ and retain the solution exactly when it satisfies all 24 inequalities. Discard repeated solutions. Singular subsets contribute no candidate. This procedure terminates because its loops and all its rational linear systems have fixed finite size.

It is complete for this intersection. At a vertex of a full-dimensional polytope, the active spatial wall coefficients span $\mathbb{R}^4$: otherwise a nonzero vector orthogonal to them permits sufficiently small perturbations in both directions while every inactive inequality remains strict, contradicting extremality. Some four active walls therefore have rank four and occur in the enumeration. Conversely a feasible solution of four independent supporting equalities is a vertex, since any segment through it in the intersection would have both endpoints in those four equalities. The intersection here is bounded, as follows directly from $|x_i|+|x_j|\le 1$, and contains the origin strictly. The exact calculation returns precisely the eight signed coordinate vertices and the sixteen half-sign vertices displayed above. Each has Euclidean norm one, so all are ideal.

Reflect across $n=(1,1,1,0,0)$ by $x\mapsto x-2\langle x,n\rangle n$. Remove the gluing wall from both cells and identify coincident walls. The resulting 38 halfspaces contain the union. Indeed, for an old wall normal $m\neq n$ and $x\in\mathcal{C}$,

$$\langle x, s_n(m)\rangle = \langle x,m\rangle - 2\langle m,n\rangle\langle x,n\rangle \le 0,$$

since both $\langle m,n\rangle$ and $\langle x,n\rangle$ are nonpositive. The mirrored containment follows by reflection. Conversely, on each side of the gluing hyperplane, their inequalities together with that side's gluing inequality include all inequalities of the corresponding original cell. Thus the new intersection equals the union. In particular no unexamined escape directions or additional vertices are being discarded.

The interiors are disjoint, so the volume is twice $4\pi^2/3$. The double needs no assumption that a saved vertex list is exhaustive. The preceding halfspace argument identifies it with the convex

union of the two cells. In the Klein model that union is the convex hull of the two finite vertex sets. Reflection and normalization give 42 distinct points on the ideal sphere, with six shared points. Strict convexity of the sphere makes every one of these points an extreme point of their hull. Hence this is the complete vertex set.

To reconstruct faces, first record, for each wall, the subset of these vertices on its equality hyperplane. Start with the full vertex set and repeatedly intersect known sets with these facet sets, discarding the empty set and repetitions, until no new set appears. There are at most 2^{42} subsets, so the procedure halts. Every generated set is the vertex set of a face, and every face of a polytope is an intersection of its containing facets. The closure is therefore complete. Its dimension is the affine rank of its vertices, computed over $\mathbb{Q}$. The resulting exact incidence counts are those in the following table; they are model checks, not a classification of other ideal right-angled four-polytopes.

Quantity	24-cell	Double
Vertices	24	42
Edges	96	168
Ridges	96	164
Facets	24	38
Triangular ridges	96	152
Quadrilateral ridges	0	12
Octahedral facets	24	30
Doubled-octahedron facets	0	8
Ridge defect W	0	12
Reflection-orbifold Euler characteristic	1	2

Each of the eight doubled-octahedron facets has nine vertices, eighteen edges, eight triangles and three quadrilaterals. There are six shared cusps and 36 other cusps. Every cusp has six incident facets, eight edges and twelve ridges. The facet–vertex check is $30 \cdot 6 + 8 \cdot 9 = 6 \cdot 42$. This completes attainment and the construction of the equality model.

A Finite data and verification scope

The universal gap proof uses the published volume formula, the antiprism bound and the stated nine-vertex classification. The equality-spectrum proof adds only the ten-vertex Figure 7 entry and three-dimensional rigidity; it proves the cuboctahedral alternative and excludes it by its cusp widths. The general chain proof does not assume a four-dimensional minimum classification: it constructs actual sections, proves their disjointness, and recognizes every all-octahedral piece by propagation in the explicit 24-cell model.

The finite supporting data have narrower purposes. The reflected double certificate checks exact rational walls, vertices, ridges, facets and cusp links. The equality certificate checks the small incidence transcriptions, the cuboctahedral model and cusp widths, the simple four-regular six-vertex graph possibilities, and cube normal identities. The D_4 root checker verifies the finite 24-cell root/vertex and reflected-octahedron facts used in the opposition argument. None of these finite model checks replaces opposite-cusp injection, the equality reductions, the surface proof, or convexity and actual-face arguments.

The finite checks use only explicitly bounded sets. The original 24-cell test is the rational active-wall algorithm just proved; the face test is the terminating facet-intersection closure. For the cuboctahedral model, rescale the spatial coordinates so the Lorentz form is $-x_0^2+(x_1^2+x_2^2+x_3^2)/2$. Its

normals are then $(1, \pm 2e_i)$ and $(1, \pm 1, \pm 1, \pm 1)$. The same vertex argument runs over the $\binom{14}{3} = 364$ triples of these fourteen rational wall equations. For the six-cusp graph check, a four-regular graph on six vertices has a perfect matching as its complement. Enumerate the $\binom{15}{3}$ triples of possible missing edges and retain those covering each vertex once. These are exactly the fifteen labelled matchings; each complement has the eight triangles and cube dual described in the proof. The cube-normal check is rational row reduction of its six square equations in eight unknowns, with kernel spanned by the constant and three sign characters. These procedures halt by their explicit finite bounds. Their completeness concerns these specified models and graph classes, not the cited classification of three-polyhedra.

The existing independent saved-data checks return the two face vectors $(24, 96, 96, 24)$ and $(42, 168, 164, 38)$, the cuboctahedron's twelve cusps with squared aspect ratio two, the fifteen labelled six-cusp graphs and cube-equation rank four. The separate D_4 check examines all $\binom{24}{2} = 276$ unordered root pairs: their spatial products $-2, -1, 0, 1$ occur respectively $12, 96, 72, 96$ times. It also checks the 96 common-neighbor cases and their $1,152$ reflected octahedral-vertex incidences. These are the recorded outcomes of the fixed checks. The universal injection, surfaces, cuts and gluing remain the written arguments above.

The source maps and solver-free commands are recorded in `docs/result_manuscript_correspondence.md` and `manuscripts/catalogue.json`. The older separate sources are preserved proof inputs. Conditional Lean interfaces distinguish the published minimum, the internal defect partition, the independent equality-spectrum reduction, the one-cut geometry and final gluing. They do not supply an end-to-end formal proof of the hyperbolic geometry by assuming the final volume or decomposition conclusion.

References

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